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$$\begin{aligned} f(x) &= \frac{\sqrt{x+7} - 3}{x-2} \\ \lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{x-2} &= \frac{\sqrt{9} - 3}{0} = \frac{0}{0} = \infty \\ &= \lim_{x \rightarrow 2} \frac{(\sqrt{x+7} - 3)(\sqrt{x+7} + 3)}{(x-2)(\sqrt{x+7} + 3)} \\ &= \lim_{x \rightarrow 2} \frac{x+7-9}{(x-2)(\sqrt{x+7} + 3)} \\ &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x+7} + 3} = \frac{1}{6} \end{aligned}$$

References